MCO-1D ORNITHOPTER

Complete Bill of Materials

Every component required to construct, wire, and commission the MCO-1D flapping-wing aircraft, down to individual fasteners and wire gauges.

Specifications derived from confirmed library constants and hardware commissioning documents.

All quantities for one complete aircraft unless noted. Revision 6 — USB flash drive added for knowledge transfer. All other changes software-only.

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SECTION 1 — AIRFRAME AND MECHANICAL STRUCTURE

All dimensions are confirmed from library constants. Wing semi-spans: W1=772mm, W2=749mm, W3=735mm, W4=712mm. Chord: 165mm uniform. Fore-aft stagger between pairs: 138mm. Vertical stagger (staircase): 100mm per step. Undercarriage height (wing root pivot to ground): 200mm.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Carbon fibre spar — main (per wing)**	3mm OD × 772mm (W1 length)	16	pcs	2 spars per wing × 8 wings. Cut to length: W1=772, W2=749, W3=735, W4=712mm. Pultruded CF rod, minimum 40GPa stiffness.
2	**Carbon fibre spar — trailing edge**	1.5mm OD CF rod	8	pcs	1 per wing. Holds membrane taut along trailing edge. Cut to wing chord length.
3	**Wing membrane material**	Ripstop nylon, 40 denier, or Mylar film 12 micron	0.6	m²	Total wing area = 0.979 m². Add 20% for trimming. Ripstop preferred for durability. Mylar for lower weight.
4	**Fuselage spine**	CF tube 10mm OD × 4mm ID, total length ~700mm	1	pcs	Main structural member. All wing root joints, ODrives, and Pi mount to this. Length sized to fit 4 wing pairs at 138mm stagger spacing.
5	**Fuselage cross-members**	CF tube 6mm OD × 4mm ID	4	pcs	One per wing pair, perpendicular to spine. Approximately 200mm each. Carry wing root joints on each side.
6	**Sector wheels**	Custom machined or 3D printed, 200° arc, radius ~35mm	8	pcs	1 per wing. 200° total travel (±100° from centre). Must withstand rope tension forces. Aluminium preferred; PETG/ABS for prototyping. Groove for 1mm rope.
7	**Wing root pivot bearings**	MR84ZZ 4×8×3mm miniature ball bearing	16	pcs	2 per wing root joint (flap axis). Must be low-friction — any drag directly reduces efficiency.
8	**Sweep pivot bearings**	MR74ZZ 4×7×2.5mm or similar	8	pcs	1 per wing for sweep (forward/aft fold) pivot. Smaller than flap bearings.
9	**Drive rope**	Spectra/Dyneema 1mm, minimum 50kg breaking strength	4	m	Connects motor shaft capstan to sector wheel. 7:1 rope drive ratio. Spectra has near-zero stretch — essential for position control accuracy. Allow 0.5m per wing pair.
10	**Motor shaft capstans**	Turned aluminium, 8mm bore, ~10mm diameter	8	pcs	1 per motor. Rope wraps around this to drive the sector wheel. Must be secured to motor shaft with grub screw.

#	Description	Part / Spec	Qty	Unit	Notes
11	**Landing gear — main struts**	CF tube 6mm OD or aluminium wire 4mm bent to shape	2	pcs	Main wheels, one each side. Height to give 200mm from wing root pivot to ground. Lightweight — this is dead weight.
12	**Landing gear — tail strut**	CF rod 3mm OD, bent	1	pcs	Tail wheel support. Must prevent tail-strike during rotation.
13	**Wheels**	Lightweight foam or foam-filled rubber, 30-40mm diameter	3	pcs	2 main + 1 tail. Keep as light as possible. Model aircraft wheels suitable.
14	**Wing attachment hardware**	M2 × 8mm stainless steel screws and M2 nuts	40	sets	Securing wing hinges, servo mounts, and cross-members. Buy extra — easy to lose.
15	**Thread-lock adhesive**	Loctite 243 (medium strength)	1	bottle	Apply to all screws that must not vibrate loose during flight. Essential for motor mount screws.
16	**Epoxy adhesive**	5-minute epoxy, structural grade	1	pack	For bonding CF components and membrane to spars. West System or similar preferred.
17	**Dihedral control linkage rods**	M2 pushrod, 50mm, with ball-link ends	8	pcs	Connect sweep servo arm to wing root to control dihedral angle. Adjustable length preferred.

■ All carbon fibre work generates fine dust. Wear an N95 mask and work outdoors or with ventilation. CF dust is a lung irritant.

SECTION 2 — FLAP MOTORS

Confirmed specification from library: GoolRC 5010 360KV brushless outrunner motors. 8 required (one per wing). These are the primary thrust and lift producers — the equivalent of the engine on a conventional aircraft.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Brushless outrunner motor**	GoolRC 5010 360KV	8	pcs	Confirmed hardware. 360KV = 360 RPM/V. At 14.8V nominal: max 5,328 RPM. With 7:1 rope drive gives adequate sector wheel speed for 3Hz flapping. Stator diameter: 50mm. Stack: 10mm.
2	**Motor mount plates**	Aluminium 2mm, custom cut to fit fuselage cross-members	8	pcs	Secure each motor to the fuselage cross-member. Must be rigid — any flex corrupts position control. Design to match motor bolt pattern.
3	**Motor mount screws**	M3 × 8mm stainless, with spring washers	32	pcs	4 screws per motor. Apply Loctite 243. Check tightness before every flight.
4	**Motor phase wires (extension)**	18AWG silicone wire, 3-conductor, red/yellow/black	2	m	If motor phase wires are too short to reach ODrive, extend. Keep as short as practical — resistance matters.
5	**Motor phase connectors**	3.5mm gold bullet connectors, male and female pairs	24	pairs	3 per motor × 8 motors = 24 pairs. For connecting motor phase wires to ODrive output. Solder carefully — poor joints cause overheating.
6	**Hall effect sensor cable**	6-pin JST-PH, 150mm	8	pcs	GoolRC 5010 has hall sensors for commutation. The connector from the motor must reach the ODrive encoder input. Extension cable if needed.

■ The GoolRC 5010 has no built-in winding temperature sensor. Motor thermal management relies on ODrive FET temperature monitoring and visual inspection. Allow motors to cool between extended test runs.

SECTION 3 — MOTOR CONTROLLERS (ODRIVES)

4 ODrive units required, each controlling 2 motors (one per channel). All communicate via CAN-FD bus. Critical setting: vel_limit = 150 rad/s on every axis.

#	Description	Part / Spec	Qty	Unit	Notes
1	**ODrive motor controller**	ODrive v3.6 (56V version) or ODrive S1	4	pcs	Each unit has 2 motor channels. 4 units × 2 channels = 8 motors total. Use 56V version for thermal headroom even though we run at 14.8V. CAN-FD bus communication.
2	**ODrive power cable**	XT30 male to bare wire, 16AWG, 150mm	4	pcs	Power input to each ODrive from PDB. 16AWG handles up to 20A continuous per ODrive.
3	**ODrive CAN bus cable**	Twisted pair 24AWG, CAT5 ethernet suitable	1	m	Daisy-chains all 4 ODrives and the Pi CAN interface. Cut from a short ethernet cable for pre-twisted pairs.
4	**CAN termination resistors**	120 ohm, 1/4W, ±1%	2	pcs	One at each END of the CAN chain. Pi end and last ODrive end. Without these the bus will not communicate reliably. Do not skip.
5	**ODrive USB cable (for configuration)**	USB-A to USB-C or Micro-USB (check ODrive model)	1	pcs	Needed during initial setup only. Connect ODrive to laptop to run ODrive Tool configuration. Not needed during flight.
6	**Heat sink compound**	Arctic Silver 5 or similar	1	tube	Apply between ODrive PCB and any heat sinking structure. Improves thermal dissipation during sustained operation.
7	**ODrive mounting standoffs**	M3 × 10mm brass standoffs	16	pcs	4 per ODrive for secure PCB mounting. Provides airflow underneath the board.

■ ODrive vel_limit MUST be set to 150 rad/s per axis before any flight testing. Factory default is much lower and will prevent the wings from reaching operating frequency. See Section 8 of the Commissioning Manual.

SECTION 4 — PITCH SERVOS

Confirmed specification: Helidirect TORQ TCM-1001. 8 required. Operate at 7.4V (from dedicated 2S LiPo or BEC), 333Hz PWM signal from Raspberry Pi GPIO. Speed: 1034°/s at 7.4V. Torque: 2.6 kg·cm at 7.4V. Running current per servo: approximately 180mA.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Pitch servo**	Helidirect TORQ TCM-1001 digital servo	8	pcs	One per wing. Controls angle of attack (AoA) of each wing. Must be fast: ≥1200°/s at operating voltage. TCM-1001 provides 1034°/s at 7.4V — confirmed adequate. DO NOT SUBSTITUTE without recalculating.
2	**Servo power BEC (7.4V)**	2S LiPo 7.4V, 500mAh OR DC-DC BEC 14.8V→7.4V, 3A	1	pcs	8 servos × 180mA = 1.44A typical. Add 100% margin: 3A BEC minimum. Dedicated 2S LiPo (500mAh) provides clean power with no switching noise. If using BEC, use LC filter.
3	**Pitch servo extension leads**	JR/Futaba servo extension, 150mm, 3-wire	8	pcs	Extends servo lead from wing to fuselage wire run. Keep short to minimise weight.
4	**Servo mounting screws**	M2 × 6mm self-tapping, with servo grommets	32	pcs	4 per servo. Always use servo grommets (rubber mounts) to isolate vibration from servo body.
5	**Servo horns / arms**	Single-arm servo horn, compatible with TCM-1001 spline	8	pcs	One per servo. Check spline count matches TCM-1001 before ordering. Metal preferred for longevity.
6	**Servo linkage ball ends**	M2 ball-link, push-fit	16	pcs	2 per wing servo (one each end of the AoA control rod). Allows angular freedom at both ends.
7	**AoA control rods**	M2 threaded steel rod, 40mm	8	pcs	One per wing. Connects servo horn to wing root pitch pivot. Adjustable thread ends for neutral angle setting.
8	**Pitch servo PWM filter capacitors**	100µF electrolytic, 16V	8	pcs	Across VCC/GND of each servo connector at the GPIO header. Prevents PWM switching noise from corrupting Pi GPIO voltage.

■ The 333Hz PWM frequency for pitch servos is non-standard (most servos use 50Hz). Verify the TCM-1001 specification confirms 333Hz compatibility before use. The library is configured for this frequency and it cannot be changed without recalibrating the servo response.

SECTION 5 — SWEEP SERVOS

8 required. DS3225-class (25kg metal gear digital servo). Operate at 50Hz PWM, 5-6V supply. Sweep range: 0° (perpendicular) to 210° (fully stowed). Library SWEEP_LIMIT_MAX_DEG = 210° at full DS3225 range.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Sweep servo**	DS3225 or equivalent 25kg metal gear digital servo	8	pcs	One per wing. Controls sweep angle (forward/aft fold of wing). 25kg·cm torque needed for wing structural loads during flight. Metal gear essential — plastic gears strip under load.
2	**Sweep servo power BEC (5-6V)**	DC-DC BEC 14.8V→5V or 6V, minimum 4A output	1	pcs	8 servos × max 500mA stall = 4A peak. 5V for standard DS3225, 6V for higher torque if supported. Check servo datasheet for max voltage.
3	**Sweep servo extension leads**	JR/Futaba servo extension, 200mm, 3-wire	8	pcs	Same as pitch servos but slightly longer as sweep servos are closer to wing roots.
4	**Sweep servo mounts**	Custom bracket, aluminium 2mm or 3D printed nylon	8	pcs	Secure sweep servo to fuselage cross-member. Must prevent servo rotation under load.
5	**Sweep servo horns**	Double-arm or wheel horn, DS3225 spline compatible	8	pcs	Wheel horn preferred — provides symmetric force on both sweep directions. Check spline size.
6	**Sweep linkage rods**	M3 threaded rod, 60mm, with M3 ball ends	8	pcs	Connects servo to wing sweep pivot. M3 for sweep (larger than pitch) due to higher forces.

SECTION 6 — RASPBERRY PI FLIGHT COMPUTER

The Raspberry Pi 4 is the main flight computer running the Wave Library. GPIO pins provide PWM to 16 servos. CAN interface connects to 4 ODrives. Serial UART connects to the Pixhawk.

#	Description	Part / Spec	Qty	Unit	Notes
1	**USB-A Flash Drive**	8 GB minimum, any brand	1	pcs	Required for knowledge transfer between aircraft. After each flight session the post-flight sequence automatically exports all discovered modes and validated principles to mco_knowledge_NNN.json. Plug the drive into the Pi USB port and run: knowledge_export_usb from the ground station. Drive then carries the knowledge file to any future MCO aircraft. Standard USB-A connection. Recommend keeping one permanently with the ground kit. No special format required — FAT32 or exFAT both work.
1	**Official Raspberry Pi 7-inch DSI Touchscreen**	Official Foundation product	1	pcs	Ground panel touchscreen. Connects via DSI ribbon connector and GPIO power — zero driver installation. Capacitive multi-touch, 800x480, 70mm x 130mm active area. Runs mco_panel.py (4 buttons: PRE-FLIGHT, POST-FLIGHT, STORM STOW, EMERGENCY STOP). Install with: sudo apt install python3-pygame
1	**Raspberry Pi 4 Model B**	4GB RAM version recommended	1	pcs	Main flight computer. 4GB RAM gives headroom for the 33,472-line Wave Library, 9 optimizer loops, 32-dimension autonomous discovery, 4-layer exploration system (biomimetic, evolutionary, random safety-gated), and knowledge transfer. 2GB works but leaves less margin.

#	Description	Part / Spec	Qty	Unit	Notes
2	**MicroSD card — HIGH ENDURANCE type**	Samsung PRO Endurance 64GB or SanDisk High Endurance 64GB. NOT the standard EVO/Ultra variants.	1	pcs	CRITICAL SELECTION: The Wave Library logs flight data every 20ms — 50 small random writes per second continuously. A standard "Class 10" card sustains 10MB/s sequential but degrades to under 1MB/s for random small writes. This can cause tick overruns and flight control jitter. The PRO Endurance and High Endurance cards are specifically rated for dashcam/CCTV duty — identical write pattern to flight logging. They maintain >4MB/s random write performance indefinitely. Do not substitute a standard card.
3	**Pi official USB-C power supply**	5V 3A USB-C, official or quality equivalent	1	pcs	ONLY for ground testing. In flight use a 5V BEC from the main battery. The Pi is very sensitive to voltage drops — a bad supply causes random crashes and corrupted SD cards.
4	**5V BEC for flight**	DC-DC step-down, 14.8V→5V, minimum 3A, USB-C output or direct 5V rail	1	pcs	Powers Pi from main LiPo during flight. Must be a quality regulated supply. Holybro or Matek BECs are reliable.
5	**CAN hat for Raspberry Pi**	MCP2515 CAN Bus Shield (SPI interface)	1	pcs	Plugs onto Pi GPIO header. Provides CAN-FD interface to the 4 ODrives. SPI bus connection to Pi. Must enable SPI in raspi-config.
6	**Pi GPIO breakout board**	40-pin ribbon cable + breadboard breakout	1	pcs	Makes servo wiring to GPIO pins manageable. Route 16 servo signal wires from breakout to servo rail.
7	**Pi case / mounting plate**	Lightweight open-frame case or bare aluminium mounting plate	1	pcs	Protects Pi from vibration damage while allowing airflow for cooling. Full enclosures trap heat — use open frame or passive heatsink.
8	**Pi heatsink set**	Adhesive copper heatsinks for CPU, RAM, USB chip	1	set	The Pi 4 runs warm under load. Heatsinks prevent thermal throttling during long test sessions. Three-piece set covers all hot chips.
9	**GPIO ribbon cable**	40-pin rainbow ribbon, 150mm	1	pcs	Connects Pi GPIO to breakout board. Keep short to minimise interference.

#	Description	Part / Spec	Qty	Unit	Notes
10	**Serial-to-UART level shifter**	3.3V/5V bi-directional level converter, 4-channel	1	pcs	For Pi serial UART to Pixhawk if Pixhawk runs at 5V logic. The Pi GPIO is 3.3V. Check Pixhawk datasheet — many are 3.3V tolerant and do not need the shifter.

SECTION 7 — PIXHAWK AUTOPILOT

The Pixhawk provides the IMU (attitude), barometric altimeter, GPS, and runs ArduPlane firmware. Communicates with the Pi at 921600 baud via MAVLink on the TELEM2 port.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Pixhawk autopilot**	Pixhawk 4 Mini, Pixhawk 6C, or Cube Orange (all compatible)	1	pcs	Any ArduPilot-compatible Pixhawk variant. Cube Orange preferred for triple-redundant IMU. Pixhawk 4 Mini is lightest option.
2	**GPS module**	Here3 or M8N GPS with compass, with mast	1	pcs	Mount on a mast above the fuselage for best sky view. Integrated compass needed for heading hold. Keep away from motor wiring (magnetic interference).
3	**Pitot tube (airspeed sensor)**	Differential pressure sensor, MS4525DO or equivalent	1	pcs	Measures airspeed. The Wave Library uses V_ms for back-side detection (6.3 m/s threshold). Pitot faces forward, away from wing downwash.
4	**Pixhawk power module**	PM07 or compatible, 3S-6S, 90A rated	1	pcs	Provides 5V to Pixhawk and measures battery voltage and current. Connects between battery and PDB.
5	**Telemetry radio (optional)**	SiK 915MHz telemetry pair, or RFD900	1	pair	For extended-range ground station. Not required for tethered TC-02/TC-03 tests where ethernet cable is used. Useful for free flight monitoring.
6	**RC receiver**	Compatible with your transmitter (SBUS, CRSF, or PPM output)	1	pcs	Receives pilot stick and lever commands. Connect to Pixhawk RC input. Use SBUS or CRSF for best latency.
7	**RC transmitter**	Minimum 6-channel, FrSky, Radiomaster, or Spektrum	1	pcs	Pilot control device. Needs at least: Left stick (pitch/yaw), Right stick (roll/collective lever), lever slider, plus 2 switches for mode selection.
8	**TELEM2 serial cable**	6-pin JST-GH to 3-pin Dupont (TX, RX, GND)	1	pcs	Connects Pixhawk TELEM2 port to Pi serial UART. Pin 2=TX, Pin 3=RX, Pin 6=GND. TX and RX cross (Pixhawk TX → Pi RX, Pixhawk RX → Pi TX).

SECTION 8 — LOAD CELLS AND TEST EQUIPMENT

Required for TC-02 (static) and TC-03 (dynamic) thrust tests. The HX711 ADC reads the load cell. Pi GPIO pins: DT=GPIO26, SCK=GPIO7.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Load cell (bar type)**	50N or 100N rated, 4-wire strain gauge	1	pcs	Measures upward thrust. Aircraft weight is 33.8N — use 50N minimum, 100N recommended for margin. Bar-type (S-beam or single-point) preferred for clean axial loading.
2	**HX711 ADC amplifier board**	HX711 24-bit ADC module	1	pcs	Amplifies and digitises load cell signal. Very common, inexpensive. RATE pin: set HIGH for 80Hz sample rate (needed for per-stroke averaging). Wires: DT→GPIO26, SCK→GPIO7, VCC→3.3V, GND→GND.
3	**Load cell calibration weight**	Known mass: 1kg and 2kg calibration weights	1	set	Needed to calibrate the load cell before TC-02. Place known weight, record reading, set scale factor. Must be accurately known mass.
4	**Test rig structure**	Steel or heavy timber frame, minimum 10kg mass	1	pcs	Must be rigid and heavy enough that vibrating wings do not move it. Weld or bolt — do not use wood screws alone. Secure to floor if possible.
5	**Tether attachment hardware**	M8 eyebolt + M8 nut + nylon lock nut	2	sets	Safety tether attachment points. Even during static testing, tether the aircraft in case a rig failure allows unexpected movement.
6	**Safety tether rope**	Dyneema 3mm, 500kg breaking, 2m length	1	pcs	Tethers aircraft to rig during TC-02. Long enough to allow full test but short enough to prevent impact if rig fails.
7	**Wing clearance guards**	Soft foam padding on rig uprights within wing arc	1	set	In case a wing strikes the rig structure during a test, foam prevents damage. The wing tips travel in a 96° arc at speed.
8	**Digital multimeter**	Any quality DMM with DC voltage and continuity	1	pcs	Essential for verifying wiring before power-on. Check all connections before first test.
9	**Bench power supply**	Variable DC, 0-20V, 5A minimum	1	pcs	For ODrive configuration and motor calibration (set to 14.8V, limit 2A). Safer than battery for bench work — current limiting prevents damage.

SECTION 9 — POWER SYSTEM

The main battery is a 4S LiPo at nominally 14.8V. Power distribution feeds: ODrives (direct), pitch servos (7.4V), sweep servos (5-6V), Raspberry Pi (5V), and Pixhawk (5V via PM).

#	Description	Part / Spec	Qty	Unit	Notes
1	**Main flight battery**	4S LiPo, 14.8V nominal, 4000-6000mAh, 45C or higher	1	pcs	Primary power source. 45C rating means it can deliver 45 × capacity in amps — a 5000mAh 45C delivers 225A peak (far more than needed). Tattu, Turnigy, or CNHL are reliable brands. NEVER use budget no-name cells.
2	**LiPo battery charger**	Balance charger, 4S compatible, minimum 50W (e.g. ISDT Q6 Plus)	1	pcs	MUST be a balance charger — charges each cell individually to prevent imbalance. Never charge LiPo with a basic DC power supply.
3	**LiPo safe charging bag**	LiPo fire-retardant bag, large enough for battery	1	pcs	Always charge inside this bag. If battery catches fire during charge the bag contains it. Not optional.
4	**Power distribution board (PDB)**	XT60 input, multiple XT30/XT60 outputs, integrated BECs preferred	1	pcs	Distributes battery power to all consumers. Look for a PDB with integrated 5V and 12V BEC outputs to reduce component count. Holybro or Matek options suitable.
5	**Main battery connector**	XT60 female (battery side) and XT60 male (PDB side)	1	pair	XT60 is the standard high-current connector for this power level. Solder with care — poor solder joints cause resistance heating and failure.
6	**Anti-spark connector**	XT60 with built-in resistor or separate XT90-S	1	pcs	Prevents the large spark when connecting the battery. Spark can damage ODrive capacitors and is a fire risk near LiPo cells.
7	**ODrive power cables**	XT30 male to XT30 female, 16AWG, 100mm	4	pcs	One per ODrive from PDB. 16AWG handles 20A continuous — more than one ODrive channel needs. Keep short to minimise resistance.
8	**5V BEC for Raspberry Pi**	DC-DC step-down, 14.8V→5V, 3A min, USB-C output preferred	1	pcs	Dedicated supply for Pi only. Do not share with servos — servo current spikes will crash the Pi. Holybro or Matek recommended.

#	Description	Part / Spec	Qty	Unit	Notes
9	**7.4V BEC for pitch servos**	DC-DC step-down 14.8V→7.4V, 3A minimum	1	pcs	8 pitch servos at 180mA each = 1.44A typical. 3A for margin. Or use a dedicated 2S LiPo (500mAh) for completely clean power.
10	**5-6V BEC for sweep servos**	DC-DC step-down 14.8V→5V, 4A minimum	1	pcs	8 sweep servos at 500mA stall each. Size for peak, not average. Keep separate from pitch servo supply for noise isolation.
11	**Battery voltage alarm**	LiPo cell voltage alarm, 2S-6S compatible	1	pcs	Plugs into balance connector and beeps loudly when any cell drops below a set threshold (set to 3.6V). Mandatory for outdoor flight.
12	**Wiring harness fuse**	ANL fuse 30A with holder, or blade fuse 30A in-line	1	pcs	In-line fuse between battery and PDB. Protects wiring in case of short circuit. Without this a short can cause a LiPo fire.

SECTION 10 — WIRING, CONNECTORS AND FASTENERS

#	Description	Part / Spec	Qty	Unit	Notes
1	**Silicone wire 18AWG**	Red and black, silicone insulation, multi-strand	3	m	Power runs: battery to PDB, PDB to ODrives. Silicone insulation is flexible and heat resistant — do not use PVC wire near ESCs.
2	**Silicone wire 22AWG**	Multiple colours	2	m	Signal wiring: servo extensions, sensor connections, CAN bus. Lighter gauge reduces weight.
3	**Silicone wire 24AWG**	Red, black, white, green for load cell	0.5	m	HX711 to Pi wiring. Short runs, low current.
4	**Heat shrink tubing**	Assorted sizes 1mm-8mm, pack of 200	1	pack	Over all soldered joints. Use correct size — too large does not shrink enough to seal.
5	**Cable ties (small)**	100mm × 2.5mm nylon, pack of 100	1	pack	Securing wire runs to fuselage. Keep all wires away from wing flap paths.
6	**Servo connector pins**	JR/Futaba female 3-pin servo connectors, pack of 20	1	pack	For custom-length servo leads or any connector that needs replacement.
7	**JST-PH connectors**	2-pin and 3-pin JST-PH male/female pairs	10	pairs	For small signal connections: hall sensor extensions, HX711 to Pi.
8	**Dupont connectors and pins**	2.54mm pitch female housing and pins, assorted	1	kit	For Pi GPIO connections. Crimp tool needed. Or use pre-made jumper wires.
9	**Pre-made jumper wires**	Female-to-female 200mm, pack of 40 in multiple colours	1	pack	Simpler alternative to crimped Dupont connectors for Pi GPIO connections. More reliable than hand-crimped for prototyping.
10	**Electrical tape**	Self-amalgamating rubber tape, 19mm wide	1	roll	Self-amalgamating (not standard PVC) for waterproofing any joint that may get wet.
11	**Solder**	60/40 rosin-core solder, 0.8mm diameter	1	reel	For all electrical connections. Lead-free is available but harder to work with — 60/40 gives better joints for beginners.
12	**Flux pen**	No-clean flux pen	1	pcs	Dramatically improves solder joints on motor phase connections and XT60 connectors. Clean flux type does not require washing after.

SECTION 11 — CAN BUS HARDWARE

The CAN-FD bus connects the Raspberry Pi to all four ODrives. All devices share a single pair of wires in a daisy-chain. Both ends of the chain MUST have 120Ω termination resistors.

#	Description	Part / Spec	Qty	Unit	Notes
1	**MCP2515 CAN hat for Raspberry Pi**	MCP2515 SPI CAN Bus module with TJA1050 transceiver	1	pcs	Plugs onto Pi 40-pin header. Provides CAN interface via SPI. Set CAN speed to 1 Mbit/s. Enable SPI in raspi-config first. Install: sudo apt install can-utils.
2	**CAN bus cable**	CAT5e ethernet cable or twisted pair 24AWG	0.5	m	Cut a short ethernet cable and use one twisted pair (e.g. orange/white-orange) for CAN-H and CAN-L. Pre-twisted pairs are essential for noise rejection.
3	**CAN termination resistors**	120Ω ±1%, 1/4W, through-hole	2	pcs	One at Pi end of CAN chain, one at last ODrive end. Solder across CAN-H and CAN-L at each termination point. Do not omit.
4	**CAN connector (board edge)**	2-pin screw terminal or JST connector for CAN-H/CAN-L	5	pcs	1 per ODrive + 1 for Pi hat. Screw terminals simplest for field maintenance.

■ To test the CAN bus before connecting hardware: sudo ip link set can0 up type can bitrate 1000000 then: cansend can0 001#DEADBEEF and: candump can0. If you see the message echoed, the bus is working.

SECTION 12 — LAPTOP AND GROUND STATION

#	Description	Part / Spec	Qty	Unit	Notes
1	**Laptop computer**	Any Mac, Linux, or Windows laptop with Python 3.8+	1	pcs	Runs the ground station client (mco_ground_station.py). No special performance needed — the application is lightweight. Screen size: larger is easier to read live data.
2	**USB-to-ethernet adapter**	USB-A or USB-C to gigabit ethernet (if laptop has no port)	1	pcs	Necessary if laptop has no built-in ethernet port. Any quality brand works. Avoid very cheap adapters — they can drop packets.
3	**Ethernet cable**	Cat5e or Cat6, 3m	1	pcs	Direct Pi-to-laptop connection. Any standard ethernet cable. No crossover needed — modern adapters auto-negotiate.
4	**Python 3.8 or later**	From python.org or system package manager	1	install	Required to run the ground station client. No additional Python packages needed — the ground station uses only the standard library. Revision 3 note: all new capabilities (HSF dynamic sweep, progressive stowing, autonomous discovery, Loop 9 meta-optimizer) are pure software additions. No new hardware is required beyond what is listed in this document.
5	**ODrive Tool (for configuration)**	pip install odrive	1	install	Needed once for ODrive configuration. Run on the laptop connected via USB to each ODrive. Not needed after initial setup.
6	**Mission Planner or QGroundControl**	Free download from ardupilot.org or qgroundcontrol.com	1	install	For Pixhawk configuration and firmware update. Windows only for Mission Planner; QGC is cross-platform. Only needed for initial Pixhawk setup.

SECTION 13 — CONSUMABLES, TOOLS AND SAFETY EQUIPMENT

#	Description	Part / Spec	Qty	Unit	Notes
1	**Soldering iron**	60W temperature-controlled, e.g. Hakko FX-888D or TS100 portable	1	pcs	Temperature control essential. Set 320-370°C for 60/40 solder. Chisel tip for XT60, fine tip for signal wiring.
2	**Wire strippers**	Automatic, 0.2-6mm range	1	pcs	For preparing all wire ends. Automatic type strips to exact depth without nicking strands.
3	**Crimping tool**	Dupont/JST crimping tool, ratchet type	1	pcs	For making custom servo lead and sensor connector ends. Ratchet type ensures consistent crimp force.
4	**Drill and drill bits**	Cordless drill + bits 2mm, 3mm, 4mm, 6mm	1	set	For motor mount holes, fuselage attachment points, and wire routing holes.
5	**Hex driver set**	Metric hex keys 1.5mm through 4mm	1	set	For M2, M3 grub screws and motor mounting. Ball-end type allows angle access.
6	**Digital calipers**	150mm digital vernier calipers	1	pcs	Measuring wing dimensions, checking fit of mechanical parts, verifying gear clearances.
7	**Scale/balance**	Kitchen scale, 5kg, 1g resolution	1	pcs	Weighing the completed aircraft for MTOW confirmation. MTOW confirmed at 3.448 kg.
8	**Safety glasses**	ANSI Z87 rated impact protection	2	pcs	One per person present during any motor or wing test. Carbon fibre splinters and rope failures are eye hazards.
9	**Ear protection**	30dB NRR earplugs or earmuffs	2	pcs	Motor + wing noise during TC-02 can exceed 85dB at 1m. Required for anyone within 3m of the rig.
10	**Fire extinguisher**	ABC dry powder, 1kg minimum, or Class D for metal fires	1	pcs	Keep within 3m of any LiPo charging or motor testing area. LiPo fires spread very fast. Know how to use it before testing.
11	**First aid kit**	Basic kit including burn dressings	1	pcs	For soldering burns and cuts from carbon fibre. Know location before starting work.
12	**Cable labels**	Self-adhesive labels or heat-shrink labels	1	pack	Label every wire at both ends during construction. A label now saves an hour of tracing later. Use: wing number, servo type (P=pitch, S=sweep), port/starboard.

#	Description	Part / Spec	Qty	Unit	Notes
13	**Isopropyl alcohol**	99% IPA, 250ml	1	bottle	Cleaning solder flux residue and PCB contacts before assembly. Do not use on servo gears or bearings.
14	**Thread-lock adhesive**	Loctite 243 medium (BLUE), 5ml	1	tube	Apply to all motor mount, servo mount, and landing gear screws. BLUE not RED — red requires heat to remove and may strip small screws.

SECTION 14 — COST SUMMARY

Approximate costs in USD at time of writing. Prices vary by supplier and region. Electronics prices in particular fluctuate. Budget an additional 20% for shipping, import duties, and replacement of any damaged items.

14.1 By Category

Airframe and mechanical (CF, bearings, rope, hardware)	~$120
Flap motors × 8 (GoolRC 5010 360KV)	~$160
ODrives × 4	~$480
Pitch servos × 8 (Helidirect TORQ TCM-1001)	~$240
Sweep servos × 8 (DS3225 class)	~$80
Raspberry Pi 4 + accessories	~$110
Pixhawk + GPS + airspeed	~$180
Load cell + HX711 + test rig hardware	~$60
Power system (batteries, BECs, PDB, charger)	~$120
Wiring, connectors, CAN hardware	~$40
Tools and consumables	~$120
Laptop (if not already owned)	~$0–800
[redacted]	**[redacted]**
SUBTOTAL (excluding laptop)	~$1,710
Contingency +20%	~$340
TOTAL ESTIMATE	~$2,050

■ The single largest variable cost is the ODrives. Budget options exist but are not recommended for this application — position control accuracy directly affects flight quality. The GoolRC 5010 motors and Helidirect TCM-1001 servos are confirmed hardware and should not be substituted without recalibrating the Wave Library.

14.2 Critical Items — Do Not Substitute

The following items have confirmed specifications embedded in the Wave Library. Substituting them without updating the library constants will result in incorrect flight behaviour:

GoolRC 5010 360KV motors	KV rating and pole pairs are used in CT calculations
Helidirect TORQ TCM-1001 pitch servos	Speed (1034°/s), torque (2.6 kg·cm), 333Hz operation all calibrated
4S LiPo main battery (14.8V nominal)	Voltage used in ODrive configuration and thermal monitoring

ODrive v3.6 or S1	CAN-FD protocol and parameter names differ between ODrive generations

End of MCO-1D Bill of Materials

Cross-reference with Commissioning Manual for installation details of each component.